

Climate change

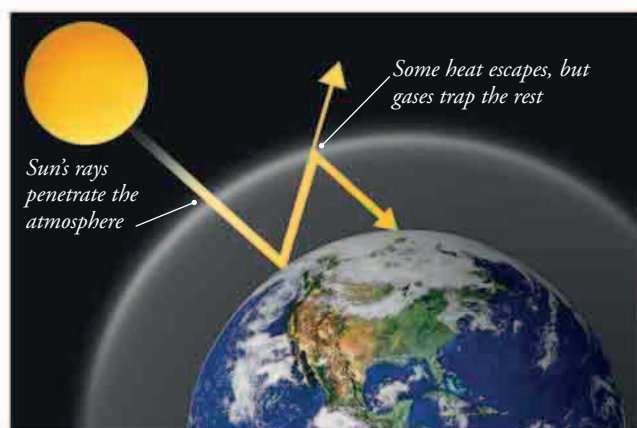
The world is warming up. Records show a steady increase in average global air temperature, which is melting polar ice and altering the pattern of the world's weather. The temperature rise is caused by a change in the nature of the atmosphere that makes it retain more heat. This change has been brought about by more than 200 years of air pollution, caused by burning coal, oil, and other fossil fuels.

WARMING WORLD

Since the 1880s, the world has warmed by an average of almost 1.8°F (1°C). It doesn't sound like much, but altogether the world has warmed by only 7°F (4°C) over the past 12,000 years, so the rate of warming is speeding up. The evidence is plain to see in the Arctic, where the area of ocean covered by ice in the late summer of 2012 was 18 percent smaller than any other September on record.

GREENHOUSE EFFECT

Certain atmospheric gases such as carbon dioxide, methane, and water vapor act like the glass of a greenhouse. They let sunshine through to heat Earth's surface but stop the heat from escaping into space. This keeps us warm—indeed, without it the planet would be too cold for life to exist. But more “greenhouse gases” in the atmosphere are boosting the greenhouse effect and raising global temperatures.



▲ BLEAK FUTURE

Polar bears live on the Arctic sea ice. As it melts away, they have nowhere to hunt, and could become extinct.

FAST FACTS

- In 2003, a heatwave in Europe killed up to 50,000 people.
- Rising ocean temperatures trigger more storms, which then cause chaos on nearby continents.
- Since 1990, 163,000 sq miles (423,000 sq km) of forest—an area the size of California—have been felled in Brazil.
- If climate change melts the Antarctic and Greenland ice sheets, sea levels could rise by up to 82 ft (25 m).